product of sine functions

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November 12, 2026

We consider the following problem:

$$f : [0; \pi]^2 \to \mathbb{R}$$

with $f(\theta, \varphi) = \sin(\theta)\sin(\varphi)$ whose landscape and level sets are illustrated as below.

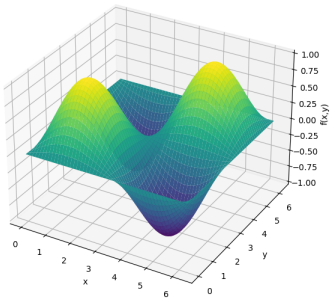


Figure 1

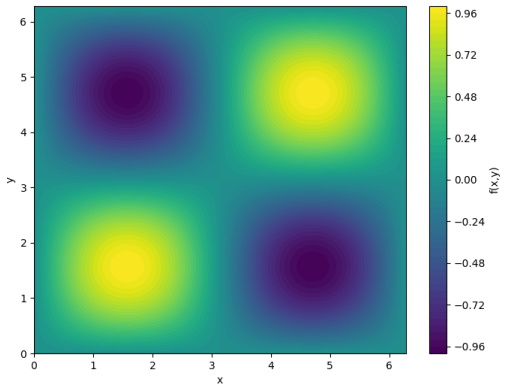


Figure 2

Observation: The function indeed admits global minimum, however there exists also several local minimums at the boundary. So numerical algorithms could get stuck at boundary if we don't have a good initialization strategy. In fact, we can specify exact regions that make gradient-based methods get stuck at local minimums.

Riemannian approach:

At the first sight, the parameter space is the product of 2 intervals $[0, \pi]$; so a potential solution to pass to Riemannian methods is using product of manifolds. Indeed, let $x \in S^1$, where S^1 is the unit sphere of $\mathbb{R}^2$, θ is the angle formed by x and x-axis, then

$$\sin(\theta) = Proj_y(x)$$

where $Proj_y(.)$ is the orthogonal projection to the y-axis. Therefore, we can define the following problem:

$$f(u, v) = Proj_y(u).Proj_y(v), \forall (u, v) \in S^1 \times S^1$$

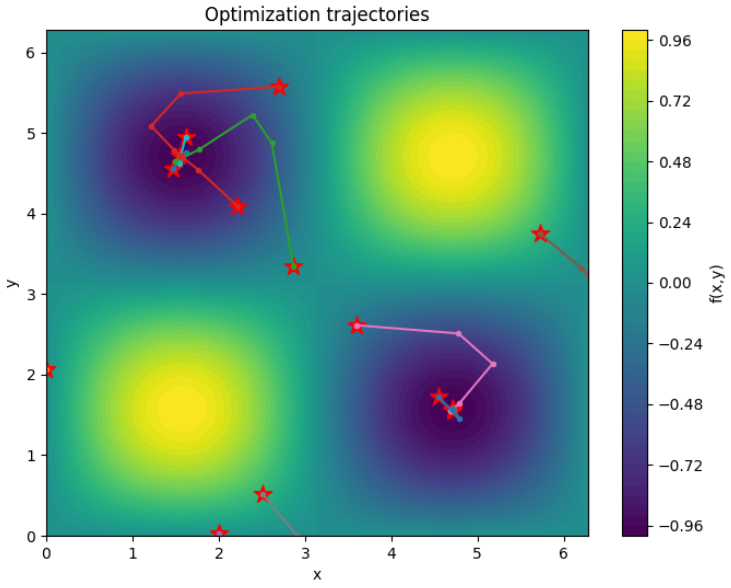


Figure 3

Product of smooth manifolds is also a manifold, f is smooth in this product, bounded from below then admits global minimums. So we continue using the first order condition optimality to compute critical points. Indeed, we have

$$\begin{aligned}\text{grad}f(u,v) &= (\text{grad}(u \to f(u,v))(u), \text{grad}(v \to f(u,v))(v)) \\ &= ((I_2 - uu^T)(0 \quad Proj_y(v))^T, (I_2 - vv^T)(0 \quad Proj_y(u))^T)\end{aligned}$$

Solving $\text{grad}f(u,v) = 0$ yields 4 critical points in which there are 2 local minimums which are also global minimum, $((0 \quad -1)^T, (0 \quad 1)^T)$ et $((0 \quad 1)^T, (0 \quad -1)^T)$

Another approach: However, we can solve this problem by a more elegant way, no need to use the product of spheres. The intuition comes from the representation of spherical coordinate, i.e. every points in R^3 can be represented by a radius r and 2 angles.

Indeed, let $x \in \mathbb{R}^3$, x can be rewritten as $(r\cos(\theta)\sin(\theta), r\sin(\theta)\sin(\theta), r\cos(\varphi))$; followed by this intuition we define the following problem:

$$f(u) = \text{Proj}_y(u), \forall u \in S^2$$

where S^2 is the unit sphere in $\mathbb{R}^3$. This problem is smooth in a compact manifold so admits global minimum. We calculate the critical points via Riemannian gradients, we have

$$\text{grad}f(u) = (I_3 - uu^T)(0 \quad 1 \quad 0)^T$$

then solving this equation yields 2 critical points $(0 \quad 1 \quad 0)$ and $(0 \quad -1 \quad 0)$; the latter is the global minimum.

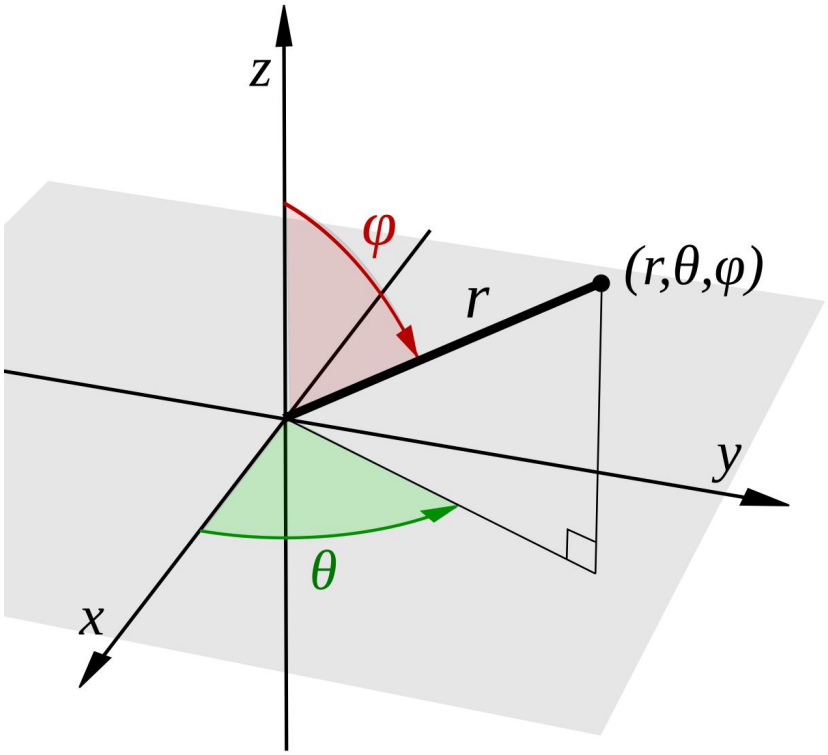


Figure 4